

JENKINS

An open-source automation server

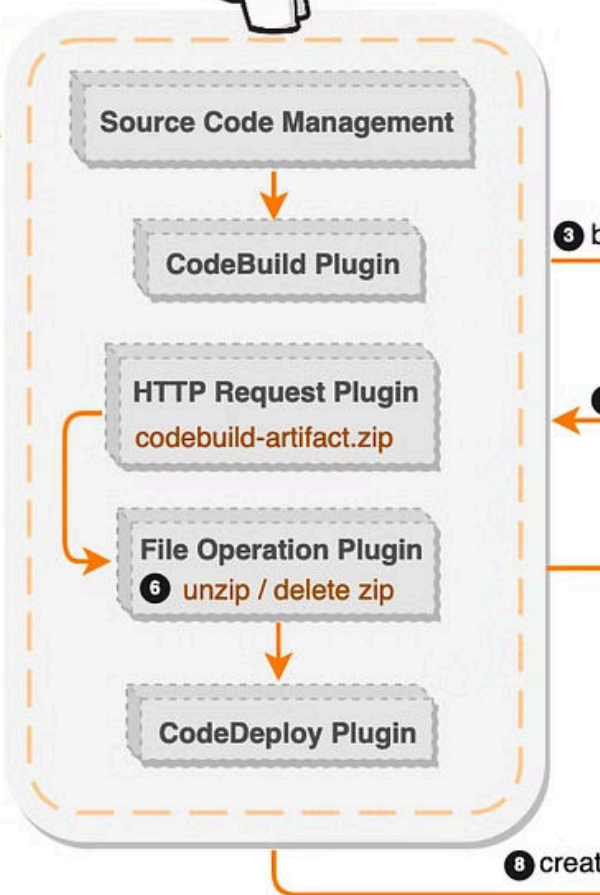
GROUP 5

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2 clone



Introduction to Jenkins

Jenkins is a software tool that helps developers automate repetitive tasks in the process of building, testing, and deploying software applications. It allows developers to integrate their code changes frequently, ensuring that software builds are reliable and consistent.

Think of Jenkins as a helpful assistant that takes care of mundane tasks like compiling code, running tests, and deploying applications, freeing up developers to focus on writing quality code and delivering value to users.

Features of Jenkins in Software Development

1 Jenkins with Plugins

You can customize Jenkins with plugins to fit your specific requirements, whether it's for testing, deployment, monitoring, or any other task in your software development process.

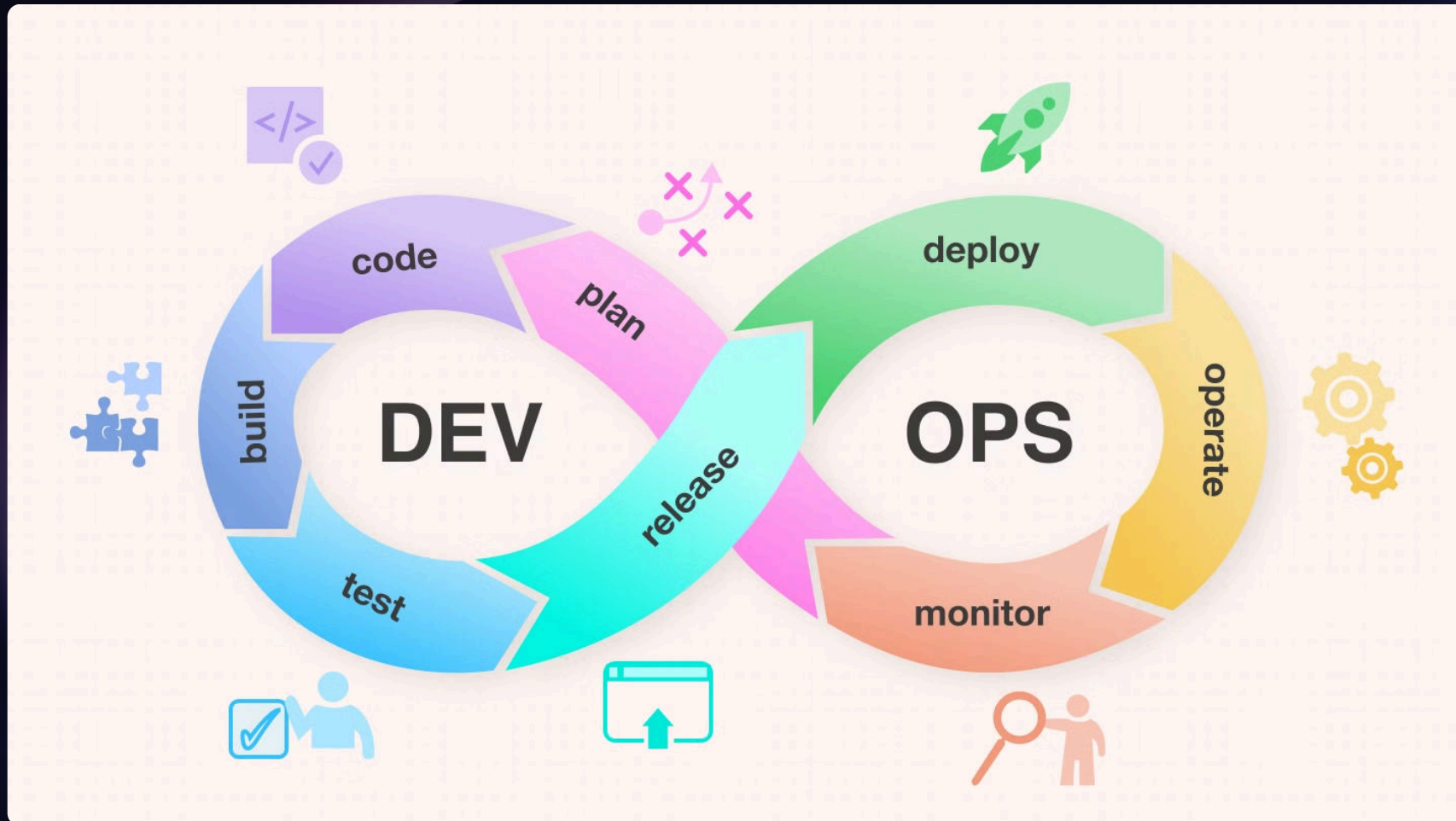
2 Automated Build and Deployment

It streamlines the process of building, testing, and deploying applications, resulting in faster and more reliable software delivery.

3 Support for Diverse Technologies

Jenkins can integrate with various tools and platforms, making it suitable for a wide range of software projects.

Jenkins in DevOps Lifecycle



Installation Options for Jenkins

- Jenkins can be installed using Jenkins installer for different operating systems or as an image in Docker or as a WAR file on any operating system.

- **Installing Jenkins**
 - Docker
 - Kubernetes
 - Linux
 - macOS
 - Windows
 - Other Systems
 - WAR file
 - Other Servlet Containers
 - Offline Installations
 - Initial Settings

- We have installed Jenkins on AWS EC2 instance using Bitnami Jenkins package.

Job and Build

Job: In Jenkins, a job is a task or unit of work that defines a series of steps to be executed, such as compiling code, running tests, and deploying applications.

- Example: "A job in Jenkins could be to build and deploy a Java application to a Tomcat server."

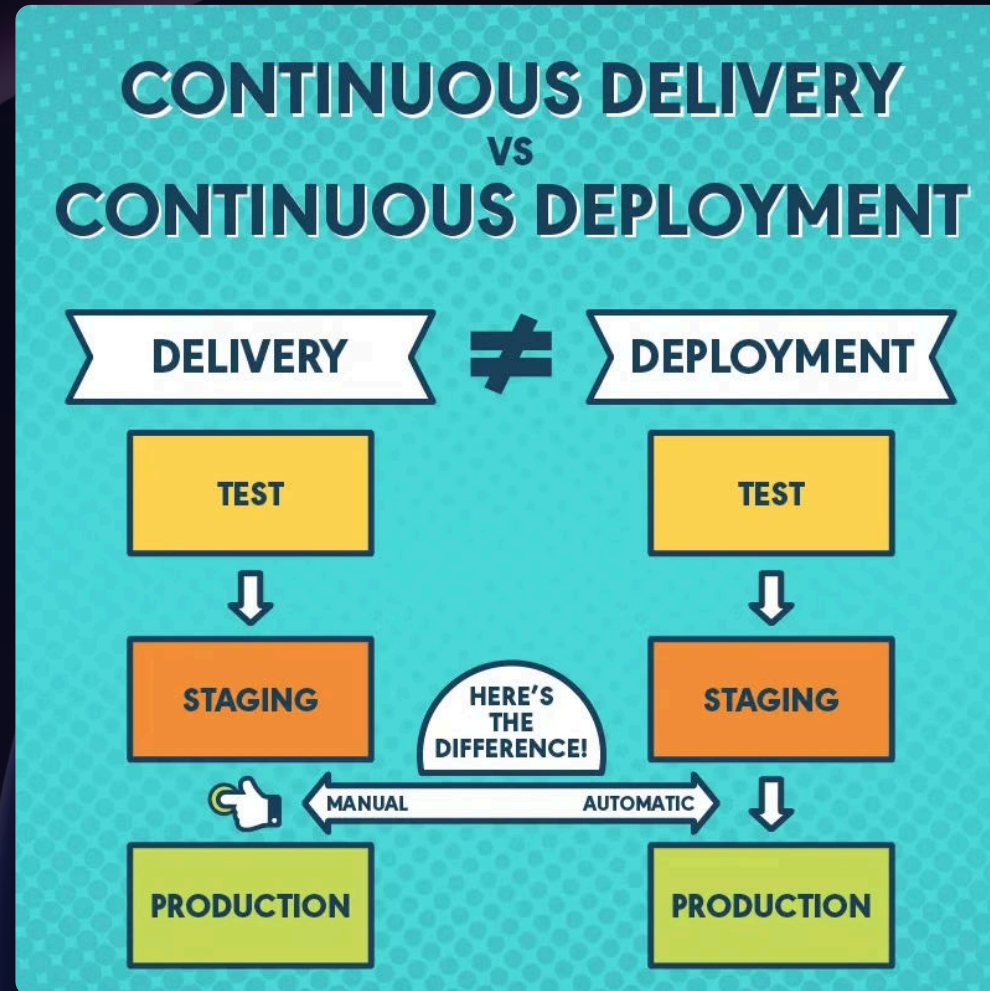
Build: A build in Jenkins refers to the process of executing the steps defined in a job, which typically involves compiling source code, running tests, and packaging the application for deployment.

- Example: "During the build process, Jenkins compiles the Java code, runs unit tests, and creates a deployable WAR file for the application."

Demonstration

[Working_Demo.mp4](#)

Delivery vs Deployment



Demonstration

[Pipeline_Demo.mp4](#)

THANK YOU